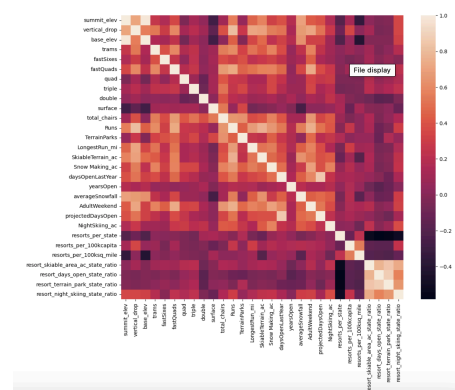


Problem Statement

Data Wrangling

Exploratory Data Analysis

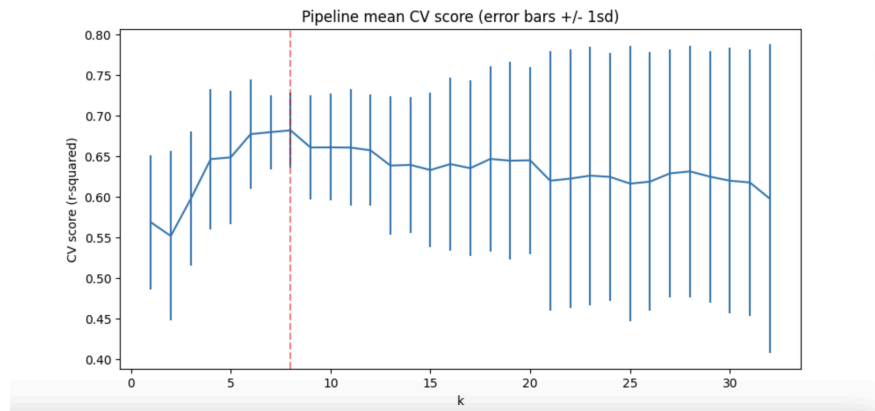
A scatter plot showing the relationship between 'vertical_drop' (x-axis) and 'Ticket price' (y-axis). The x-axis ranges from 0 to 4500, and the y-axis ranges from 0 to 150. The data points are blue circles. There is a general upward trend, with ticket prices increasing as vertical drop increases, though with significant variance. The density of points is highest between 0 and 2000 vertical drop and 50 to 100 ticket price.



Data Preprocessing and Feature Engineering

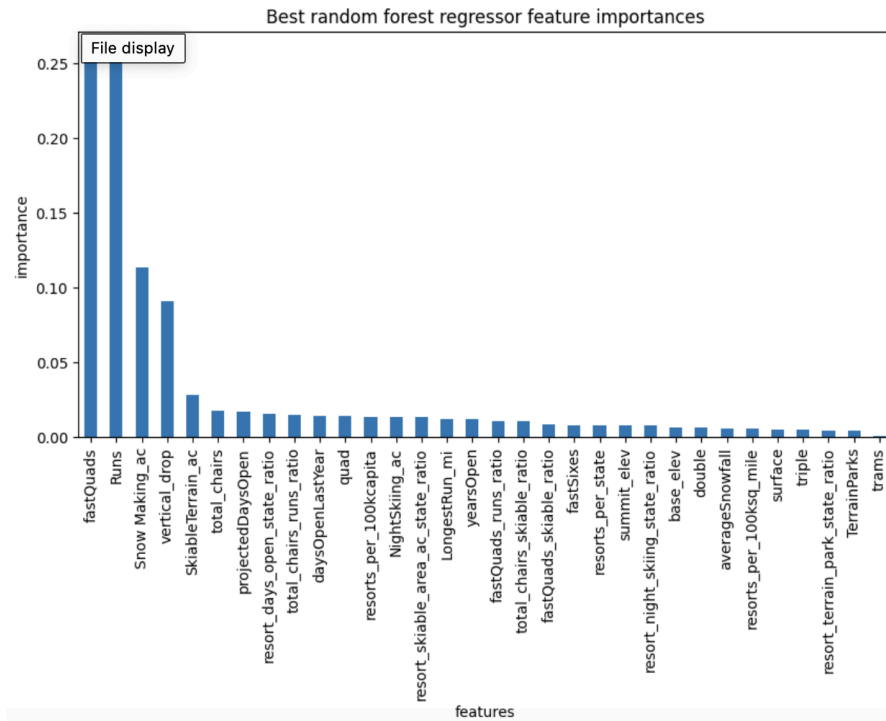
Feature selection was performed using SelectKBest to identify the most important predictors.

This reduced noise and improved model performance by focusing on the most relevant features.



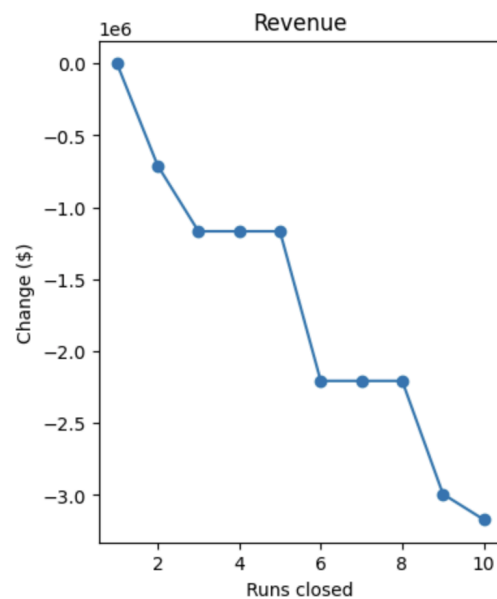
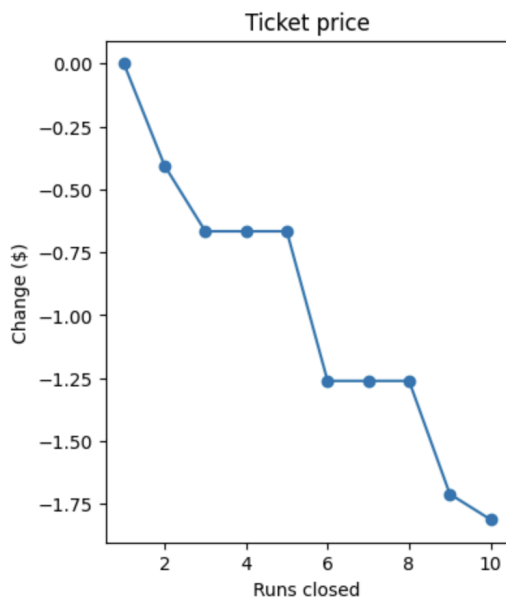
Model Evaluation and Development

Linear regression and Random Forest Regressor were tested using cross-validation and R^2 score to ensure reliability. The Random Forest model was selected as the final model because it outperformed linear regression and captured non-linear relationships more effectively.



Scenario Modeling and Analysis

Scenario modeling was used to evaluate how changes to resort features would affect pricing and revenue.



Pricing Recommendation

Big Mountain currently charges \$81 per ticket, while the model suggests it could support a price closer to \$96. This indicates the resort may be underpricing itself. Rather than immediately increasing prices, a gradual increase is recommended to monitor customer response and demand. The best case scenario involved adding 1 run, increasing vertical drop by 150 feet, adding 1 chair lift. This scenario increased the supported ticket price by approximately \$1.99 per ticket and could lead to increased revenue. However, this should only be pursued if the operating cost per ticket of the new chair lift is lower than this increase. Other scenarios, such as increasing snowmaking or extending the longest run, did not produce meaningful improvements and are not recommended. Run closures should be approached carefully. While closing one run has minimal impact, larger closures reduce both ticket price and revenue. Any closures should be tested gradually using low-traffic runs and monitored closely.

Conclusion

The analysis shows that Big Mountain is likely underpriced relative to its facilities. The Random Forest model provided strong predictive performance and helped identify the most important factors influencing ticket price. Scenario modeling provided actionable insights into how operational changes affect revenue and pricing.

Future Scope of Work

This analysis was limited by the available data. The dataset did not include detailed operating costs, customer demand and behavior, competition and pricing strategy, or weather conditions. Including these factors would improve the model's accuracy and usefulness. To make the model

more practical for business use, it could be deployed as a dashboard or web application, allowing analysts to test different scenarios without needing a data scientist each time.